



CERTIFICATE OF TESTS

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Printed: 05/16/2024

Heat: 2137105
Material: 5L- X52 4-1/2X.337 (15.00#)BR,DRL,PEB (11000102)
Vendor: Nucor Hickman - Armorel AR US (180436)

Purchase Order: 26632645
Sales Order: 1001000690
MFG Date: 03/14/2024

CHEMICAL PROPERTIES

	C	MN	P	S	NI	CR	CU	MO	SI	TI	V	CE
Ladle	0.0400	1.0600	0.0120	0.0014	0.0300	0.0400	0.0900	0.0100	0.2100	0.0120	0.0050	0.1082
	Al: 0.0300		Sn: 0.0042		N: 0.0060		B: 0.0000		Ca: 0.0010		Nb: 0.0330	
Chem	0.0546	1.0863	0.0109	0.0017	0.0306	0.0350	0.0690	0.0067	0.2184	0.0111	0.0051	0.1238
	Al: 0.0284		Sn: 0.0038		N: 0.0222		B: 0.0002		Ca: 0.0014		Nb: 0.0346	
Chem	0.0531	1.0834	0.0106	0.0015	0.0308	0.0352	0.0693	0.0067	0.2162	0.0112	0.0051	0.1221
	Al: 0.0286		Sn: 0.0038		N: 0.0148		B: 0.0002		Ca: 0.0014		Nb: 0.0347	

Acceptance Criteria: CEPCM = 0.2500

MECHANICAL PROPERTIES

YS (ksi): 70.60 TS (ksi): 74.40 EL (%) : 30.00 HRB: 88.00 Y/T(%): 0.95

CHARPY TEST RESULTS (BODY)

Orientation : TRANSVERSE	Specimen(S)	Energy (ft-lb)	%Shear
Spec. Size (in) : .394"X.197"X2.165"	1	69.00	100.00
Test Temp (°F) : -20	2	73.00	100.00
Location : 3T90	3	75.00	100.00
	Average	72.33	100.00

Acceptance Criteria Min. (ft-lb) : 10

COMMENTS

MINIMUM HYDROSTATIC TEST PRESSURE (PSI) : 4,700 PER 5 SECONDS

MANUFACTURED TO THE 46TH EDITION OF API 5L SPECIFICATION
MANUFACTURED IN THE USA-HIGH FREQUENCY WELDED PRODUCT
WELD SEAM ANNEALING AT 1600 DEGREES F.MINIMUM
ULTRASONIC TESTED TO N-10 NOTCH 1/8 DRILLED HOLE
FLATTENING TEST PASSED
1 INCH L90 STRIP SPECIMEN
THIS MATERIAL MEETS :ASTM A53B-22,ASME SA53B-21 LATEST EDITION
API 5L PSL 2 GRADE X52M
MEETS HARDNESS REQUIREMENTS OF NACE MRO 175/ISO 15156-2, CLAUSE A.2.1.1
MEETS ISO 10474 3.1.B/EN 10204: 3.1 / STANDARDS ARE LATEST REVISION
MELTED AND MANUFACTURED IN THE U.S.A.
STEEL SUPPLIER: Nucor Hickman - Armorel AR US
COIL MANUFACTURER: Nucor Hickman - Armorel AR US

The above analyses and tensile properties are correct to the best of my knowledge and belief.

Quality Control Dept